

Telematics tracker - parts purchase list (5 boards + spares)

From the verified BOM (tag v1.0.4-mfg-2026-09-30, 77 lines). "Ask for" is the plain description to give a shop; the part number is the exact part the design was checked with; the substitute column says when another brand or part is acceptable. Buy quantities include spares. Passives are usually sold in strips of 10/50/100.

Local SMD shop / Robu / Evelta - 43 lines

Buy	Ask for	Part number (manufacturer)	LCSC	Substitute allowed?	Designators	Per board
12	U.FL / IPEX MHF1 SMD RF receptacle (2 mm)	XY-IPEX1 XYECONN	C53133524	Any brand U.FL (IPEX1 / MHF1) receptacle	AF1,AF2	2
20	SMD ceramic capacitor 0603, 18pF, 50 V, C0G	CL10C180JB8NNNC Samsung Electro- Mechanics	C1647	Any brand, same dielectric (C0G/NP0 required)	C1,C2	2
20	SMD ceramic capacitor 0805, 4.7uF, 25 V, X7R	CC0805KKX7R8BB475 YAGEO	C354262	Any brand, must be X7R (X5R/Y5V not accepted), >= 25 V	C12,C63	2
20	SMD ceramic capacitor 0603, 4.7nF, 50 V, X7R	CL10B472KB8NNNC Samsung Electro- Mechanics	C1621	Any brand, must be X7R (X5R/Y5V not accepted), >= 50 V	C22	1
20	SMD ceramic capacitor 0603, 6.8pF, 50 V, C0G	CL10C6R8CB8NNNC Samsung Electro- Mechanics	C318672	Any brand, same dielectric (C0G/NP0 required)	C3,C4	2
170	SMD ceramic capacitor 0603, 100nF, 50 V, X7R	CC0603KRX7R9BB104 YAGEO	C14663	Any brand, must be X7R (X5R/Y5V not accepted), >= 50 V	C5,C6,C8,C9,C10,C11,C13,C20,C23,C24,C25,C26,C27,C28,C29,C43,C45,C46,C47,C72,C76,C80	22
20	SMD ceramic capacitor 0603, 47nF, 50 V, X7R	CL10B473KB8NNNC Samsung Electro- Mechanics	C1622	Any brand, must be X7R (X5R/Y5V not accepted), >= 50 V	C64	1
70	SMD ceramic capacitor 0603, 1uF, 25 V, X7R	CC0603KRX7R8BB105 YAGEO	C106858	Any brand, must be X7R (X5R/Y5V not accepted), >= 25 V	C7,C14,C21,C42,C44,C61,C68,C69,C75	9
20	SMD ceramic capacitor 0402, 100nF, 16 V, X7R	CL05B104KO5NNNC Samsung Electro- Mechanics	C1525	Any brand, must be X7R (X5R/Y5V not accepted), >= 16 V	C83	1
20	SMD ceramic capacitor 0402, 100pF, 50 V, C0G	0402CG101J500NT FH	C1546	Any brand, same dielectric (C0G/NP0 required)	C84	1

20	SMD ceramic capacitor 0402, 10nF, 50 V, X7R	CL05B103KB5NNNC Samsung Electro-Mechanics	C15195	Any brand, must be X7R (X5R/Y5V not accepted), >= 50 V	C85	1
6	Rectifier diode S3M, 3 A 1000 V, SMB (DO-214AA)	S3M TWGMC	C5204901	Any brand S3M	D1	1
12	Dual diode BAV99 (series pair), SOT-23	BAV99,215 Nexperia	C2500	Any brand BAV99	D10,D11	2
6	TVS diode SMF5.0A, 5 V unidirectional, SOD-123FL	SMF5.0A MDD(Microdiode Semiconductor)	C193402	Any brand SMF5.0A	D12	1
6	SMD LED 0603 red	KT-0603R Hubei KENTO Elec	C2286	Any brand 0603 red LED	D13	1
6	SMD LED 0603 green	KT-0603G Hubei KENTO Elec	C12624	Any brand 0603 green LED	D20	1
6	SMD LED 0603 blue	KT-0603B Hubei KENTO Elec	C2288	Any brand 0603 blue LED	D21	1
12	Dual diode BAV70 (common cathode), SOT-23	BAV70,215 Nexperia	C2501	Any brand BAV70	D5,D6	2
18	Schottky diode SS3200, 3 A 200 V, SMA (DO-214AC)	SS3200 MDD(Microdiode Semiconductor)	C65001	Any brand SS3200 or SS3200-class 200 V 3 A schottky in SMA. SS34 (40 V) is NOT acceptable	D7,D8,D16	3
20	Ferrite bead 0805, 600 ohm @ 100 MHz, >= 500 mA	GZ2012D601TF Sunlord	C1017	Any brand 0805 600R ferrite bead	FB1	1
6	JST XH 3-pin vertical through-hole header, 2.5 mm (B3B-XH-A) - battery connector	B3B-XH-A(LF)(SN) JST	C144394	JST B3B-XH-A or any XH-compatible 3-pin vertical header	J2	1
6	PNP transistor MMBT3906, SOT-23	MMBT3906,215 Nexperia	C75549	Any brand MMBT3906	Q13	1
6	P-channel MOSFET AO3401A, 30 V 4 A, SOT-23	AO3401A AOS	C15127	Any brand AO3401A	Q3	1
60	NPN transistor MMBT3904, SOT-23	MMBT3904(RANGE:100-300) JSCJ	C20526	Any brand MMBT3904	Q4,Q5,Q6,Q7,Q8,Q9,Q10,Q11,Q12,Q14	10

70	SMD resistor 0805, 10k ohm, 1 %	0805W8F1002T5E UNI-ROYAL	C17414	Any brand, 1 %	R1,R2,R43,R44,R47,R48,R51,R63,R87	9
50	SMD resistor 1206, 12k ohm, 1 %	1206W4F1202T5E UNI-ROYAL	C17912	Any brand, 1 %	R14,R15,R16,R17,R18,R19	6
20	SMD resistor 0805, 220k ohm, 1 %	0805W8F2203T5E UNI-ROYAL	C17556	Any brand, 1 %	R21,R22	2
20	SMD resistor 0805, 100 ohm, 1 %	0805W8F1000T5E UNI-ROYAL	C17408	Any brand, 1 %	R23,R24	2
90	SMD resistor 0805, 100k ohm, 1 %	0805W8F1003T5E UNI-ROYAL	C149504	Any brand, 1 %	R25,R26,R30,R31,R32,R33,R34,R35,R50,R62,R81	11
20	SMD resistor 0805, 0 ohm (jumper link)	0805W8F0000T5E UNI-ROYAL	C17477	Any brand, 1 %	R3,R69	2
20	SMD resistor 0805, 9.1k ohm, 1 %	0805W8F9101T5E UNI-ROYAL	C17855	Any brand, 1 %	R36,R37	2
20	SMD resistor 0805, 1M ohm, 1 %	0805W8F1004T5E UNI-ROYAL	C17514	Any brand, 1 %	R38,R39	2
20	SMD resistor 0805, 330 ohm, 1 %	0805W8F3300T5E UNI-ROYAL	C17630	Any brand, 1 %	R4,R6	2
60	SMD resistor 0805, 4.7k ohm, 1 %	0805W8F4701T5E UNI-ROYAL	C17673	Any brand, 1 %	R5,R42,R46,R53,R55,R57,R59,R64	8
50	SMD resistor 0805, 47k ohm, 1 %	0805W8F4702T5E UNI-ROYAL	C17713	Any brand, 1 %	R52,R54,R56,R58,R60,R61	6
20	SMD resistor 0805, 1k ohm, 1 %	0805W8F1001T5E UNI-ROYAL	C17513	Any brand, 1 %	R65	1
30	SMD resistor 0805, 33 ohm, 1 %	0805W8F330JT5E UNI-ROYAL	C17634	Any brand, 1 %	R66,R67,R68	3
30	SMD resistor 0805, 2.2k ohm, 1 %	0805W8F2201T5E UNI-ROYAL	C17520	Any brand, 1 %	R7,R8,R45,R49	4
20	SMD resistor 0402, 0 ohm (jumper link)	0402WGF0000TCE UNI-ROYAL	C17168	Any brand, 1 %	R71,R72	2
20	SMD resistor 0805, 4.3k ohm, 1 %	0805W8F4301T5E UNI-ROYAL	C17667	Any brand, 1 %	R83	1
20	SMD resistor 0805, 1.5k ohm, 1 %	0805W8F1501T5E UNI-ROYAL	C4310	Any brand, 1 %	R84	1
20	SMD resistor 0603, 51k ohm, 1 %	0603WAF5102T5E UNI-ROYAL	C23196	Any brand, 1 %	R91	1

7	Crystal 8 MHz, SMD 3225 (3.2 x 2.5 mm) 4-pin, load capacitance 12 pF	X32258MOB4SI YXC Crystal Oscillators	C2682775	Any brand 8 MHz 3225 with CL = 12 pF	Y1	1
---	--	---	----------	--------------------------------------	----	---

Distributor: Mouser India / DigiKey India / element14 India - 26 lines

Buy	Ask for	Part number (manufacturer)	LCSC	Substitute allowed?	Designators	Per board
30	SMD ceramic capacitor 1210, 47 uF, 10 V, X7R	GRM32ER71A476KE15L muRata	C84494	Murata/Samsung/TDK, must be X7R and 1210; X5R not accepted	C40,C41,C81,C82	4
60	SMD ceramic capacitor 0805, 10 uF, 25 V, X7R or X7S	GRM21BC71E106KE11L muRata	C109040	Any brand, X7R/X7S; X5R not accepted	C62,C65,C66,C67,C77,C78,C79	7
30	SMD ceramic capacitor 1210, 2.2 uF, 100 V, X7R	TCC1210X7R225K101MT CCTC	C5449052	Any brand, MUST be rated 100 V, X7R, 1210	C70,C71,C73,C74	4
6	ESD diode array PESD5V0L5UY, 5-line, SOT-363 (SC-70-6)	PESD5V0L5UY,115 Nexperia	C179747	Nexperia PESD5V0L5UY; else a 5-line 5 V SOT-363 array with pin 2 = GND and <= 15 pF/line	D14	1
6	USB ESD protection USBLC6-2SC6, SOT-23-6	USBLC6-2SC6 ST	C7519	ST USBLC6-2SC6 or any brand with the same name/pinout	D15	1
6	TVS diode SMDJ100A, 100 V standoff, 3000 W, SMC (DO-214AB), UNIDIRECTIONAL	SMDJ100A Littelfuse	C1977839	Littelfuse / Bourns / Vishay SMDJ100A. NOT the bidirectional SMDJ100CA	D2	1
6	CAN bus ESD/TVS diode PESD1CAN, SOT-23	PESD1CAN,215 Nexperia	C15771	Nexperia PESD1CAN or NUP2105L (onsemi)	D3	1
6	SMD fuse 1 A 250 V, Littelfuse 0443 series (Nano2, 10.1 x 3.1 mm)	0443001.DR Littelfuse	C95352	Littelfuse 0443001.DR only (footprint-specific)	F1	1
6	Micro-Fit 3.0 compatible header, 2 x 6 = 12 pins, 3.0 mm pitch, VERTICAL through-hole (the harness connector)	WAFER-MX3.0-12PZZ XUNPU	C7588012	Molex 43045 series 12-circuit vertical header, or XUNPU WAFER-MX3.0-12PZZ from LCSC	J1	1
6	Shielded power inductor 150 uH, 12.3 x 12.3 mm, Isat >= 2.7 A, DCR <= 0.3 ohm	SMDRI127-151MT Shun Xiang Nuo	C21325	Any 12 x 12 mm shielded 150 uH inductor with Isat >= 2.7 A (check the datasheet)	L1	1
6	Common-mode choke TDK ACT45B-510-2P (51 uH, CAN bus)	ACT45B-510-2P-TL003 TDK	C76584	TDK ACT45B-510-2P-TL003 only (footprint-specific)	L2	1
6	Power inductor 2.2 uH, 2.5 x 2.0 mm (1008 / 2520), >= 2.2 A	DFE252012P-2R2M=P2 muRata	C391305	Murata DFE252012P-2R2M or same-size metal-alloy 2.2 uH >= 2 A	L3	1
20	Chip inductor 68 nH 0402 (GNSS antenna bias)	LQW15AN68NG80D muRata	C3221844	Any brand 0402 68 nH wire-wound, +/-2 or 5 %	L4	1
12	4 Pin SOP Phototransistor Photocoupler (OPTO-SMD-4_L4.4-W4.1-P2.54-LS7.0-BL)	EL357N(D)(TA)-G EVERLIGHT	C42385002	Exact part	OK1,OK2	2

20	SMD resistor 0805, 60.4 ohm, 1 %	AC0805FR-0760R4L YAGEO	C228935	Any brand, 1 % - E96 value, may need a distributor	R11,R12	2
20	SMD resistor 2512, 1 ohm, 1 %, ANTI-SURGE / pulse-withstanding type, >= 1 W	FRS2512F1R00TS FOJAN	C55348540	Any brand 2512 1R 1 % anti-surge (not a plain thick film)	R80	1
20	SMD current-sense resistor 0805, 0.1 ohm, 1 %, >= 0.25 W	RL0805FR-7W0R1L YAGEO	C327057	Any brand 0805 0R1 1 %	R82	1
20	SMD resistor 0805, 976 ohm, 1 %	0805W8F9760T5E UNI-ROYAL	C17880	Any brand, 1 % - E96 value, may need a distributor	R85	1
20	SMD resistor 0805, 536 ohm, 1 %	FRC0805F5360TS FOJAN	C5713309	Any brand, 1 % - E96 value, may need a distributor	R86	1
20	SMD resistor 0805, 5.23k ohm, 1 %	FRC0805F5231TS FOJAN	C2933464	Any brand, 1 % - E96 value, may need a distributor	R88	1
20	SMD resistor 0805, 30.1k ohm, 1 %	0805W8F3012T5E UNI-ROYAL	C21841	Any brand, 1 % - E96 value, may need a distributor	R89	1
20	SMD resistor 1210, 68 ohm, 1 %, 0.5 W	RC1210FR-0768RL YAGEO	C137091	Any brand 1210 68R 1 %	R90	1
6	Battery charger TI BQ25606RGER, VQFN-24	BQ25606RGER TI	C374063	Exact part	U6	1
6	SPI NOR flash GD25Q64ESIGR 64 Mbit, SOIC-8 208 mil (5.3 mm wide), industrial grade	GD25Q64ESIGR GigaDevice Semicon Beijing	C2831359	GigaDevice GD25Q64E or Winbond W25Q64JVSSIQ (same package)	U7	1
6	Level shifter TI TXB0104PWR, TSSOP-14	TXB0104PWR TI	C60708	Exact part (TI TXB0104PWR)	U8	1
7	Crystal 32.768 kHz, SMD 3215 (3.2 x 1.5 mm) 2-pin, load capacitance 7 pF	Q13FC13500002 EPSON	C48615	Epson FC-135 / Q13FC13500002 or any 3215 32.768 kHz with CL = 7 pF	Y2	1

LCSC direct (lcsc.com, ships to India) - China-brand part - 7 lines

Buy	Ask for	Part number (manufacturer)	LCSC	Substitute allowed?	Designators	Per board
12	N-channel MOSFET AM2390N, 150 V, SOT-23	AM2390N-TP TECH PUBLIC	C51886143	Exact from LCSC preferred. Substitute only if: SOT-23, VDS >= 100 V, ID >= 1 A, RDS(on) specified at VGS 4.5 V	Q1,Q2	2
6	MCU Artery AT32F403ACGT7, LQFP-48	AT32F403ACGT7 ARTERY	C55058656	Exact part (LCSC or Artery's India distributor)	U2	1
6	IMU QST QMI8658B, LGA-14	QMI8658B QST	C5380158	Exact part (LCSC)	U3	1
6	CAN transceiver SIT1051AT/3, SOP-8, 3.3 V I/O	SIT1051AT/3 SIT	C5382551	NXP TJA1051T/3 (pin-compatible, Mouser/DigiKey) is the substitute	U4	1

6	Buck converter EG Micro EG11752, SOIC-8 with exposed pad, 100 V	EG11752 EG Micro	C53368402	Exact part, LCSC only - no substitute	U5	1
6	LDO 3.3 V 500 mA ME6211C33M5G, SOT-23-5	ME6211C33M5G-N MICRONE	C82942	Diodes AP2112K-3.3 (same SOT-23-5 pinout VIN-GND-EN-NC-VOUT) is the substitute	U9	1
6	Nano-SIM card holder, push-push, 7 pins, 1.37 mm height, SMD	NANO SIM 7P 1.37H PUSH JXTCONN	C53207808	Exact from LCSC, or another nano-SIM push-push holder ONLY if its footprint matches the drawing	X1	1

Quectel India distributor (or Mouser India) - 1 lines

Buy	Ask for	Part number (manufacturer)	LCSC	Substitute allowed?	Designators	Per board
6	LTE Cat 1 + GNSS module Quectel EC200U-CN (EC200UCNAA-N05-SGNSA), LCC-144	EC200UCNAA-N05-SGNSA Quectel	C2916205	Exact part. Buy from a Quectel India distributor or Mouser India	U1	1

Notes

- **Never substitute** without checking: U5 (EG11752), U1 (modem), U2 (MCU), U3 (IMU), U6 (charger), U8 (TXB0104), L2 (CAN choke), F1 (fuse footprint), D2 (must be the unidirectional SMDJ100A), crystals (load capacitance), the SIM holder (footprint).
- Capacitor dielectric matters: buy X7R where it says X7R (many shops offer X5R/Y5V by default - refuse them); C0G/NP0 for the crystal and RF capacitors.
- E96 resistor values (60.4R, 536R, 976R, 5.23k, 30.1k) are rarely in local shops - order those from a distributor with the rest of the "D" list.
- LCSC ships the exact C-numbers to India (courier + customs duty); it is the only source for the China-brand parts in the "C" list, and usually the cheapest for everything else.
- Assembly: the modem (LCC-144), charger (QFN-24, thermal pad) and IMU (LGA-14) need a stencil and reflow. Order the stencil with the bare boards (JLCPCB, ~\$7) and use a local SMT shop, or a hot plate/oven at home for the first boards.